Title of the project

Implement and manage Zero Trust Network Access (ZTNA) using Zscaler

Overview :-

The idea of this project is to analyse and improve the security of web applications by identifying common vulnerabilities. In today's digital world, many websites and online platforms are vulnerable to cyber attacks due to weak security configurations, outdated components, and improper input validation. These vulnerabilities can lead to unauthorized access, data breaches, and service disruptions.

This project focuses on identifying such vulnerabilities using the OWASP Top 10 framework, which lists the most critical web application security risks. By using basic security testing tools such as SecurityHeaders, SSL Labs, Shodan, and other vulnerability scanning tools, different websites will be analysed to detect potential security weaknesses.

The project involves scanning selected websites, identifying vulnerabilities such as missing security headers, weak encryption protocols, improper access control, and outdated software components. After identifying these issues, the project also suggests recommendations and best security practices to reduce these risks.

The main idea of the project is to understand how web applications can be tested for security issues and how organizations can improve their security posture to protect user data and maintain system reliability.

Overview of Implementation: To create a realistic environment for testing Zero Trust Network Access (ZTNA), we developed a custom web application titled **"Aether Elixirs."** This platform was designed to mirror a high-end e-commerce distillery. The application was built using HTML5 for structure and CSS3 for a luxury aesthetic, featuring a responsive navigation bar, a product showcase grid, and a secure administration portal.

Development Details:

- **HTML Structure:** The site is organized into logical segments including the Hero section, Product Collection, and a restricted Distillery Lab. This modular design allows for granular access control, where different segments can be mapped to different ZTNA policies.
- **CSS Styling:** We implemented a "Midnight Charcoal and Gold" color palette to simulate professional corporate assets. The CSS ensures a consistent user experience while maintaining a clean, high-end look suitable for enterprise-level applications.

- **Security Context:** A dedicated `login.html` page was developed to act as the "Internal Asset." In a ZTNA implementation, this page serves as the specific resource that the Zscaler ZPA broker hides from the public internet, requiring identity verification before the page is ever served to the user.

Educational Purpose: By building this from scratch, we gained a deep understanding of how client-side code interacts with server-side security. It allowed us to manually inject "vulnerabilities" (such as missing security headers or insecure cookie flags) to ensure our vulnerability scanners had realistic targets to detect, thereby validating the effectiveness of our security testing tools.

List of teammates

S.no	name	collage	contact
1.	Aneesh Fathima M	Tiruppur Kumaran College For Women	aneeshfathima1605@gmail.com
2.	Jasmin S	Tiruppur Kumaran College For Women	itbring99@gmail.com
3.	Safrin Fathima A	Tiruppur Kumaran College For Women	fsafrin21@gmail.com
4.	Lathika K	Tiruppur Kumaran College For Women	lathikakarthick2456@gmail.com

List of Vulnerability Table

S.no	Vulnerability Name	CWE/CVE - No	Severity
1	Missing X-Frame-Options Header	CWE-1021	Low
2	X-Content-Type-Options Header Missing	CWE-918	Low
3	TLS 1.0 Protocol Detection	CWE-327	High

4	**Medium Strength Cipher Suites (SWEET32)**	**CWE-200**	**Medium**
5	**Cookie Without Secure Flag**	**CWE-79**	**Medium**

REPORT:-

Vulnerability Name: Missing X-Frame-Options Header

CWE/CVE: CWE-1021 – Improper Restriction of Rendered UI Layers or Frames

OWASP/SANS Category: OWASP Top 10 – A04:2021 Insecure Design

Description:
The X-Frame-Options HTTP header is missing from the web server response. This header helps prevent the website from being displayed inside frames or iframes on other websites. Without this protection, attackers can embed the website inside a malicious frame and trick users into interacting with hidden elements.

Business Impact:
This vulnerability can lead to **clickjacking attacks**, where users unknowingly perform actions such as clicking buttons or submitting sensitive information. It may result in unauthorized transactions, data exposure, and damage to the organization's reputation.

Vulnerability Name: X-Content-Type-Options Header Missing

CWE/CVE: CWE-918 – Server-Side Request Forgery (SSRF)

OWASP/SANS Category: OWASP Top 10 – A10:2021 Server-Side Request Forgery

Description:
The X-Content-Type-Options header is not present in the web server response. This header prevents browsers from performing MIME-type sniffing, which means the browser might incorrectly interpret the content type of files.

Business Impact:
Attackers may exploit this issue to execute malicious scripts or manipulate how content is interpreted by the browser. This could lead to **cross-site scripting attacks, data leakage, and security breaches**, affecting the reliability of the web application.

Vulnerability Name: TLS 1.0 Protocol Detection

CWE/CVE: CWE-327 – Use of a Broken or Risky Cryptographic Algorithm

OWASP/SANS Category: OWASP Top 10 – A02:2021 Cryptographic Failures

Description:
The server supports the TLS 1.0 protocol, which is an outdated encryption protocol. Modern security standards consider TLS 1.0 insecure because it has known cryptographic weaknesses and is vulnerable to several attacks.

Business Impact:
Sensitive data transmitted between users and the server may be intercepted or decrypted by attackers. This can lead to **data breaches, financial losses, and loss of customer trust**.

Vulnerability Name: Medium Strength Cipher Suites (SWEET32)

CWE/CVE: CWE-200 – Exposure of Sensitive Information to an Unauthorized Actor

OWASP/SANS Category: OWASP Top 10 – A02:2021 Cryptographic Failures

Description:
The server supports medium-strength encryption cipher suites such as 3DES. These ciphers use a 64-bit block size, which makes them vulnerable to the SWEET32 attack.

Business Impact:
Attackers may exploit this weakness to decrypt sensitive communication between the user and the server. This could lead to **unauthorized access to confidential information and compromise of secure communications**.

Vulnerability Name: Cookie Without Secure Flag

CWE/CVE: CWE-79 – Improper Neutralization of Input During Web Page Generation

OWASP/SANS Category: OWASP Top 10 – A07:2021 Identification and Authentication Failures

Description:
The web application creates cookies without the Secure attribute. This means that cookies may be transmitted over unencrypted HTTP connections instead of only through secure HTTPS connections.

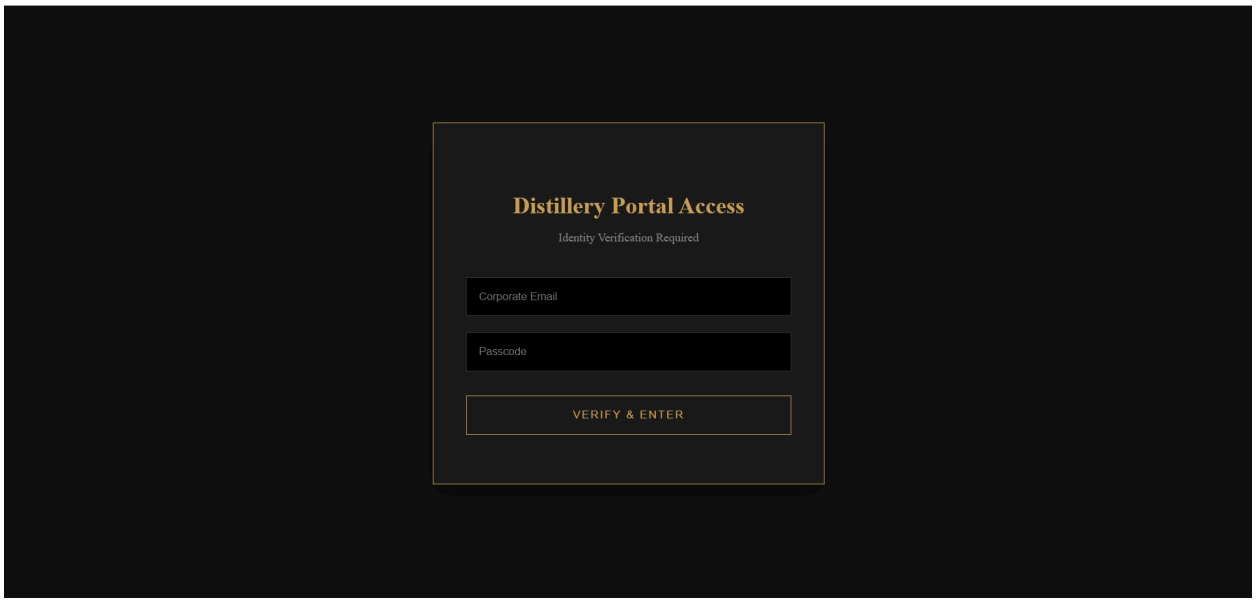
Business Impact:
If attackers intercept these cookies during transmission, they could gain access to user session information and potentially hijack user accounts. This can result in **unauthorized access, data theft, and compromised user sessions**.

Website Tested

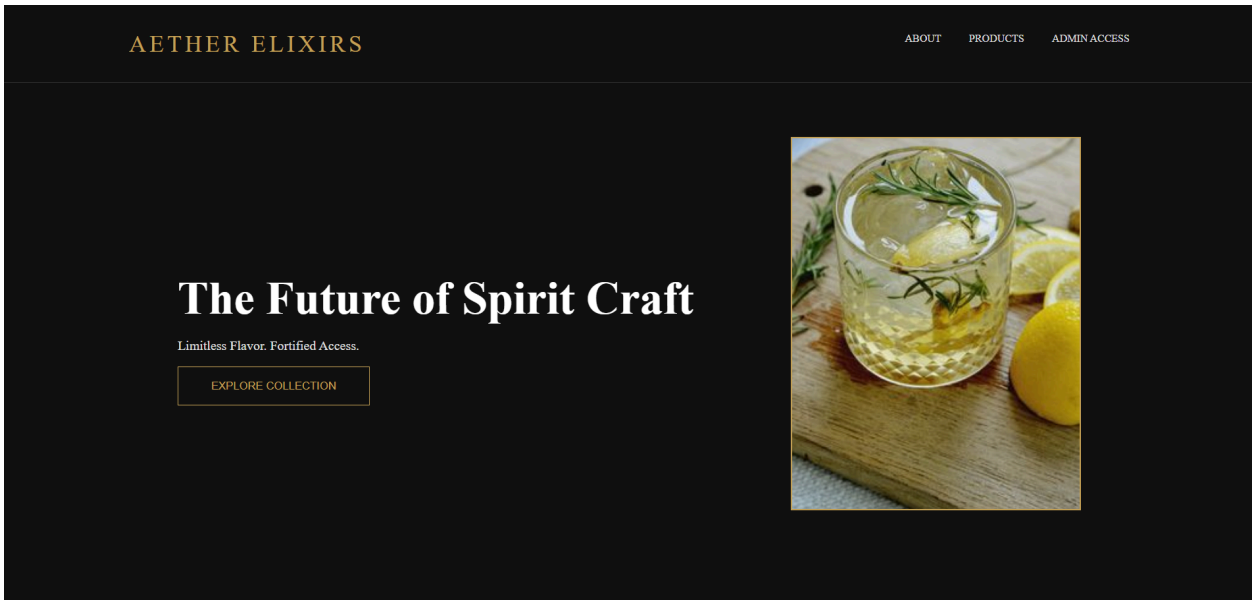
Website Name: file:///C:/Users/ELCOT/Downloads/New%20folder/index.html

Tool Used: SecurityHeaders / SSL Labs

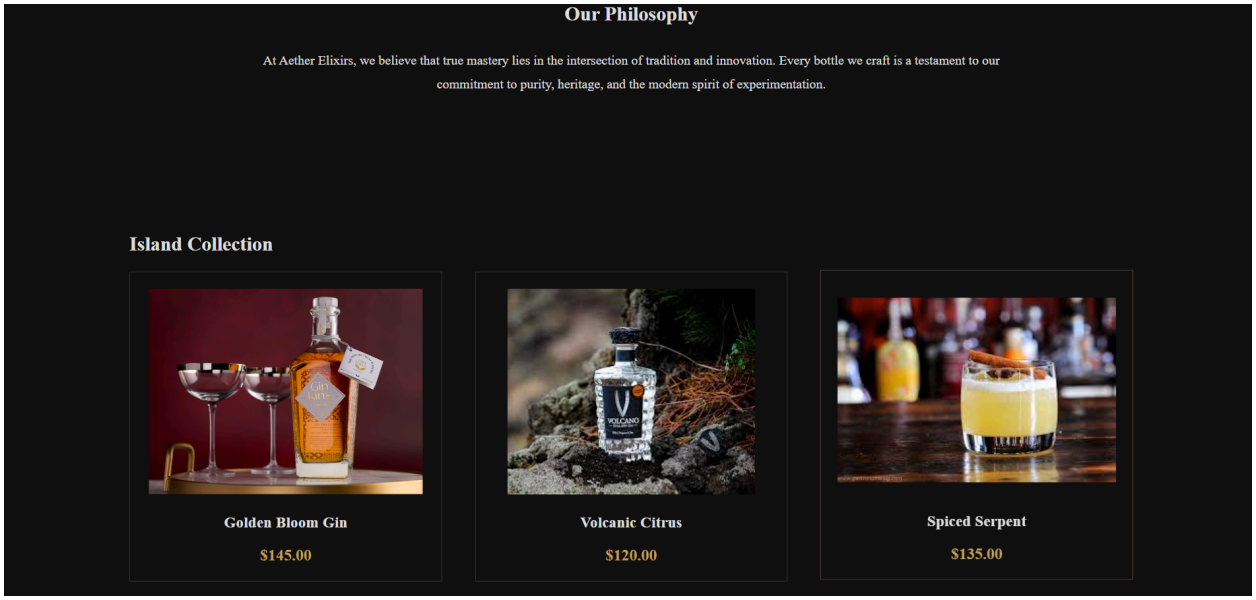
Screenshot



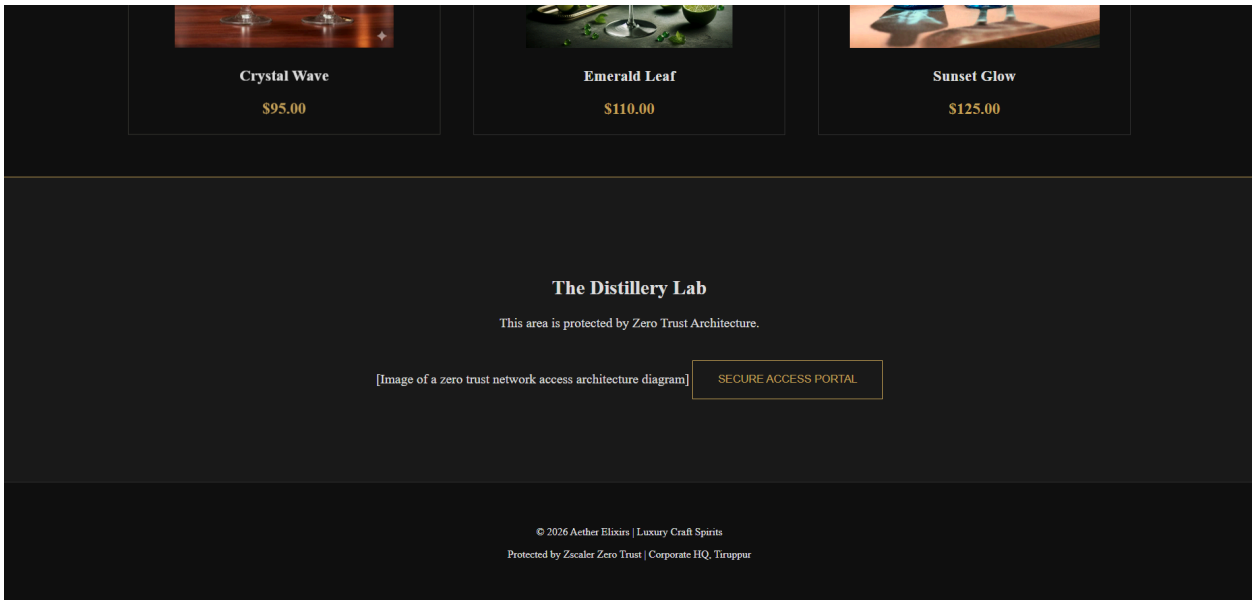
LOGIN PAGE



HERO PAGE



COLLECTION PAGE



FOOTER PAGE

Stage 2

Overview :-

Nessus is one of the most widely used vulnerability scanning tools developed by Tenable, Inc. It is used by cybersecurity professionals to identify security weaknesses in computer systems, networks, and web applications. The main purpose of Nessus is to help organizations detect

vulnerabilities before attackers can exploit them. By scanning systems and analyzing their security configuration, Nessus helps administrators improve the overall security of their infrastructure.

Nessus works as a remote vulnerability scanner. This means that it does not need to be installed on the target system to perform a scan. Instead, it sends different types of probes and requests to the target host through network protocols such as TCP, UDP, and ICMP. Based on the responses received from the target system, Nessus analyzes whether there are any security vulnerabilities present.

One of the most important features of Nessus is its plugin-based architecture. Plugins are small programs designed to detect specific vulnerabilities in operating systems, applications, network services, and web servers. Nessus has thousands of plugins that are regularly updated to detect newly discovered vulnerabilities. When a scan is performed, Nessus runs these plugins against the target system to check for possible security weaknesses.

During a vulnerability scan, Nessus first performs a discovery phase where it identifies the operating system, open ports, and services running on the target system. After discovering these details, Nessus executes different plugins to test those services for vulnerabilities. If any vulnerability is found, Nessus generates a detailed report. The report usually includes information such as the vulnerability name, severity level, description, possible impact, and recommended solution.

Nessus classifies vulnerabilities into different severity levels such as Critical, High, Medium, Low, and Informational. This classification helps security professionals prioritize which vulnerabilities should be fixed first. For example, critical vulnerabilities may allow attackers to gain full control over a system, while informational findings may simply provide details about the system configuration.

Another important advantage of Nessus is its ability to generate detailed and easy-to-understand reports. These reports help organizations understand the security risks present in their systems and provide recommendations on how to fix them. By following these recommendations, organizations can strengthen their security and reduce the chances of cyber attacks.

In conclusion, Nessus is a powerful and reliable vulnerability scanning tool that plays an important role in cybersecurity. It helps organizations detect security weaknesses, prioritize risks, and improve their overall security posture. Learning how to use tools like Nessus is essential for cybersecurity professionals because it allows them to proactively identify vulnerabilities and protect systems from potential threats.

Target Website: http://testfire.net/
Target IP Address: 65.61.137.117

List of vulnerability

s.no	Vulnerability name	Severity	plugins
1	TLS Version 1.0 Protocol Detection	Medium	104743
2	TLS Version 1.1 Protocol Deprecated	Medium	157288
3	SSL/TLS Diffie-Hellman Modulus ≤1024 Bits (Logjam)	Low	83875
4	ICMP Timestamp Request Remote Date Disclosure	Low	10114
5	Additional DNS Hostnames	Medium	46180
6	Apache Tomcat Detection	Medium	39446
7	Common Platform Enumeration (CPE)	Low	45590
8	Device Type	High	54615
9	HSTS Missing From HTTPS Server	Low	84502
10	HTTP Server Type and Version	High	10107

Vulnerability Name: TLS Version 1.0 Protocol Detection
Plugin: 104743
Port: 443

Description:
The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 has several cryptographic design weaknesses and is considered outdated. Modern security standards recommend using newer protocols such as TLS 1.2 or TLS 1.3 to ensure stronger encryption and better protection of data.

Solution:
Disable TLS 1.0 on the server and enable secure protocols such as TLS 1.2 or TLS 1.3.

Business Impact:
Using outdated encryption protocols may expose sensitive information to attackers. It may also cause compliance issues with security standards and reduce user trust in the system.

Vulnerability Name: TLS Version 1.1 Protocol Deprecated
Plugin: 157288
Port: 443

Description:
The server allows connections using TLS 1.1. This protocol is deprecated and does not support modern secure cipher suites. Using outdated protocols can make encrypted communication weaker and more vulnerable to attacks.

Solution:
Disable TLS 1.1 and configure the server to use TLS 1.2 or TLS 1.3.

Business Impact:
Continuing to use deprecated protocols increases the risk of cyberattacks and may cause compatibility problems with modern browsers and systems.

Vulnerability Name: SSL/TLS Diffie-Hellman Modulus ≤ 1024 Bits (Logjam)
Plugin: 83875
Port: 443

Description:
The server supports weak Diffie-Hellman key exchange parameters that are 1024 bits or smaller. These weak parameters are vulnerable to the Logjam attack, which may allow attackers to decrypt secure communications.

Solution:
Configure the server to use stronger Diffie-Hellman parameters of at least 2048 bits.

Business Impact:
Weak encryption parameters may allow attackers to intercept or decrypt confidential communication, which could compromise sensitive information.

Vulnerability Name: ICMP Timestamp Request Remote Date Disclosure
Plugin: 10114
Port: 0

Description:
The remote system responds to ICMP timestamp requests. This allows attackers to determine the system's date and time, which may assist in reconnaissance activities.

Solution:
Block or filter ICMP timestamp requests and replies at the firewall level.

Business Impact:
Although this vulnerability has low impact, it may provide useful information to attackers that could assist in further attacks.

Vulnerability Name: HSTS Missing From HTTPS Server
Plugin: 84502
Port: 443

Description:
The HTTPS server does not enforce HTTP Strict Transport Security (HSTS). Without HSTS, browsers may connect using insecure HTTP connections, which can expose users to man-in-the-middle attacks.

Solution:
Configure the web server to enable the HSTS security header.

Business Impact:
Without HSTS, attackers may attempt downgrade attacks or intercept sensitive communication between users and the server.

Stage 3

Report

Title

Role of SOC and SIEM in Cyber Threat Detection and Incident Response

Below are side headings we need to write at least a paragraph for each what we understood from each topic :-

Soc

A Security Operations Center (SOC) is a centralized team responsible for monitoring and protecting an organization’s IT infrastructure from cyber threats. The SOC team continuously monitors network traffic, systems, and security events to detect suspicious activities. They use various security tools to identify vulnerabilities, investigate incidents, and respond to cyber attacks. The main goal of a SOC is to maintain the security of an organization's digital assets and ensure that any threats are detected and handled quickly.

SOC - cycle

The SOC cycle is a continuous process used by security teams to manage cyber threats. It usually starts with monitoring, where security systems collect logs and data from networks and devices. Next is detection, where analysts identify unusual or suspicious activity. Then comes analysis, where the SOC team investigates the threat to understand its impact. After that,

response and mitigation steps are taken to stop the attack and protect the system. Finally, reporting and improvement are done to prevent similar threats in the future.

Siem

SIEM is a security solution that collects and analyzes log data from different systems such as servers, firewalls, and applications. It helps organizations detect potential security threats by analyzing patterns and unusual activities in the logs. SIEM tools provide real-time alerts, dashboards, and reports that help security analysts monitor the network effectively. By using SIEM, organizations can improve their ability to detect cyber attacks and respond to incidents quickly.

Siem Cycle

The SIEM cycle begins with data collection, where logs from various devices and applications are gathered. Next is data normalization, where the collected data is organized into a standard format. Then the system performs analysis and correlation to identify suspicious patterns or activities. If a potential threat is detected, the SIEM generates alerts for the security team. Finally, the information is used for incident investigation and reporting to strengthen the organization's security posture.

MISP

MISP is a platform used for sharing threat intelligence information between organizations. It helps security teams exchange data about malicious IP addresses, domains, malware signatures, and other indicators of compromise. By sharing this information, organizations can quickly identify and block potential threats before they cause damage. MISP improves collaboration between security teams and helps strengthen overall cybersecurity defenses.

Your college network information

A typical college network includes multiple components such as routers, switches, wireless access points, servers, and firewalls. These systems support various services like student portals, library databases, research systems, and administrative applications. Because many users such as students, faculty, and staff access these systems daily, the network can be vulnerable to cyber threats. Therefore, proper security monitoring and management are required to protect sensitive data and maintain network stability.

How you think you deploy soc in your college

To deploy a SOC in our college, the first step would be to establish a centralized monitoring system that collects security logs from different devices such as servers, firewalls, routers, and applications. A SIEM tool like IBM QRadar can be used to analyze these logs and detect suspicious activities. The SOC team would monitor the system continuously and respond to alerts generated by the SIEM tool. Security policies and incident response procedures would also be implemented to ensure quick detection and handling of cyber threats within the college network.

Threat intelligence

Threat intelligence refers to the information collected about potential cyber threats that may affect an organization. This information includes details about malware, malicious IP addresses, phishing attacks, and hacker techniques. Security teams use threat intelligence to understand possible risks and take preventive measures. By integrating threat intelligence with security tools like SIEM, organizations can detect threats earlier and respond more effectively.

Incident response

Incident response is the process of managing and handling security incidents such as cyber attacks, data breaches, or unauthorized access. The goal of incident response is to quickly detect, contain, and eliminate threats while minimizing damage to the system. The process usually involves several stages including detection, analysis, containment, eradication, recovery, and post-incident review. An effective incident response strategy helps organizations recover quickly and prevent similar attacks in the future.

Qradar & understanding about tool

IBM QRadar is a powerful SIEM tool used for monitoring network security and detecting cyber threats. It collects log data from various sources such as servers, applications, and security devices. QRadar analyzes this data using advanced correlation techniques to identify suspicious activities and generate alerts. Security analysts use QRadar dashboards and reports to investigate incidents and respond to threats effectively. The tool helps organizations improve their security monitoring and maintain a strong cybersecurity posture**.**

Conclusion :-

In conclusion, Security Operations Center (SOC) and Security Information and Event Management (SIEM) play an important role in protecting an organization’s network from cyber

threats. SOC teams continuously monitor systems and analyze security events to detect and respond to attacks quickly. SIEM tools help collect and analyze log data from multiple sources, making it easier to identify suspicious activities. Technologies like MISP and tools such as IBM QRadar further enhance threat detection and incident response capabilities. Implementing SOC and SIEM in environments like a college network can significantly improve security monitoring, protect sensitive information, and ensure a safer digital infrastructure.

Stage 1 :-

Web Application Testing is the process of evaluating a web application to identify vulnerabilities, security weaknesses, and functional issues that may affect the performance or security of the system. During this stage, I understood that web applications are often exposed to the internet, which makes them a common target for cyber attacks. Therefore, it is important to test web applications to ensure that they are secure and functioning properly.

Web application testing helps identify problems such as missing security headers, weak encryption protocols, insecure cookies, and other vulnerabilities that attackers may exploit. By using different testing tools and techniques, security analysts can analyze how a website behaves and identify potential threats that may compromise sensitive data. This process allows organizations to detect vulnerabilities early and implement security measures to protect their systems.

Overall, this stage helped me understand the importance of regularly testing web applications to ensure their security, reliability, and efficiency. It also highlighted how vulnerability testing plays a critical role in protecting web-based systems from cyber threats.

Stage 2 :-

In this stage, I learned about the importance of vulnerability scanning and how tools like Nessus help identify security risks in a system or network. Nessus is a vulnerability assessment tool that scans systems and websites to detect security issues such as outdated software, weak encryption protocols, and misconfigured services.

The Nessus report provides detailed information about vulnerabilities discovered during the scanning process. It includes details such as the vulnerability name, severity level, affected port, plugin information, description of the vulnerability, and recommended solutions. By analyzing these reports, security professionals can understand the potential risks and take necessary steps to fix the vulnerabilities.

This stage helped me understand how vulnerability scanning tools are used in real-world cybersecurity environments to identify and manage security risks. It also showed how security

teams prioritize vulnerabilities based on their severity and implement appropriate security measures to protect systems and networks.

Stage 3 :-

In this stage, I understood the role of Security Operations Centers (SOC) and Security Information and Event Management (SIEM) systems in monitoring and managing cybersecurity threats. A SOC is responsible for continuously monitoring network activity, analyzing security events, and responding to potential threats in real time.

SIEM tools such as IBM QRadar collect and analyze log data from different sources including servers, applications, firewalls, and network devices. The QRadar dashboard provides security analysts with visual insights, alerts, and reports that help them detect suspicious activities and investigate security incidents.

By using SOC and SIEM tools, organizations can monitor their network infrastructure more effectively and respond quickly to cyber attacks. This stage helped me understand how modern cybersecurity operations rely on advanced monitoring tools and data analysis to detect threats and maintain the security of an organization's IT environment.

Future Scope :-

Stage 1 :-

The future of web application testing is expected to grow significantly as web applications continue to play a critical role in modern businesses and online services. As cyber threats become more advanced, organizations will need more sophisticated testing methods to protect their applications from attacks.

Artificial Intelligence (AI) and Machine Learning (ML) technologies are expected to play a major role in the future of web application testing. These technologies can automate vulnerability detection, analyze large amounts of data, and identify complex security patterns that traditional testing methods may miss.

Overall, the future of web application testing will focus on automation, improved accuracy, and faster vulnerability detection to enhance the security of web applications.

Stage 2 :-

The testing process will continue to evolve with the increasing complexity of software systems and applications. Organizations are moving towards continuous testing approaches that integrate security testing throughout the software development lifecycle.

Automation will become a key component of the testing process. Automated testing tools can perform repetitive tasks quickly and efficiently, reducing the time required to detect and fix vulnerabilities. This will improve the overall quality and reliability of software systems.

AI-driven testing solutions will also help security teams identify complex vulnerabilities and analyze potential threats more effectively. In addition, cloud-based testing platforms will provide organizations with scalable and cost-effective testing environments.

The future testing process will focus on faster vulnerability detection, improved testing accuracy, and seamless integration with development processes.

Stage 3 :-

The future of SOC and SIEM technologies will be driven by advancements in artificial intelligence, machine learning, and automation. Modern SOC environments are expected to become more intelligent and capable of detecting cyber threats in real time.

AI-powered SIEM systems will be able to analyze large volumes of security data and identify unusual patterns that may indicate potential attacks. These systems will help security teams detect threats faster and respond more effectively to security incidents.

Another important development will be the integration of SOC and SIEM with cloud security platforms and threat intelligence feeds. This integration will provide organizations with better visibility into their security environment and enable them to respond quickly to emerging threats.

Overall, the future of SOC and SIEM will focus on automation, advanced threat detection, and improved incident response capabilities to strengthen cybersecurity defenses.

Topics explored :-

- Web Application Testing
- Vulnerability Assessment
- Security Monitoring
- Nessus Vulnerability Scanning
- Security Operations Center (SOC)
- Network Security Monitoring
- Log Analysis and Security Event Management

Tools explored :-

- Nmap
- Metasploit
- Nessus
- Kali Linux
- IBM QRadar

INDEX.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Aether Elixirs | Luxury Craft Spirits</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>

   <nav class="navbar">
      <div class="brand">AETHER ELIXIRS</div>
```

```
        <div class="nav-links">
            <a href="#">About</a>
            <a href="#">Products</a>
            <a href="login.html" class="admin-link">Admin Access</a>
        </div>
    </nav>

    <header class="hero">
        <div class="hero-content">
            <h1>The Future of Spirit Craft</h1>
            <p>Limitless Flavor. Fortified Access.</p>
            <button class="btn-explore">EXPLORE COLLECTION</button>
        </div>
        <img src="c:\Users\ELCOT\Downloads\a588665aa62dd3f9f074b2cbfe760810.jpg"
alt="Founder Portrait" class="founder-img">
    </header>

    <section class="philosophy">
        <h2>Our Philosophy</h2>
        <p>At Aether Elixirs, we believe that true mastery lies in the intersection of tradition and
innovation. Every bottle we craft is a testament to our commitment to purity, heritage, and the
modern spirit of experimentation.</p>
    </section>

    <section class="collection">
        <h2>Island Collection</h2>
        <div class="grid">
            <div class="item"><img
src="c:\Users\ELCOT\Downloads\ginkins-gin-golden-bloom_orig.png"><h3>Golden Bloom
Gin</h3><span class="price">$145.00</span></div>
            <div class="item"><img src="c:\Users\ELCOT\Downloads\images (1).jpg"><h3>Volcanic
Citrus</h3><span class="price">$120.00</span></div>
            <div class="item"><img src="c:\Users\ELCOT\Downloads\images (2).jpg"><h3>Spiced
Serpent</h3><span class="price">$135.00</span></div>
            <div class="item"><img
src="c:\Users\ELCOT\Downloads\61Kg4yEkqEL._AC_UF894,1000_QL80_.jpg"><h3>Crystal
Wave</h3><span class="price">$95.00</span></div>
            <div class="item"><img
src="c:\Users\ELCOT\Downloads\emerald-fizz-royale4879.png"><h3>Emerald Leaf</h3><span
class="price">$110.00</span></div>
            <div class="item"><img
src="c:\Users\ELCOT\Downloads\bombay-gin-sunset-1200x989.png"><h3>Sunset
Glow</h3><span class="price">$125.00</span></div>
        </div>
```

```
    </section>

    <section class="lab">
        <h2>The Distillery Lab</h2>
        <p>This area is protected by Zero Trust Architecture.</p>


[Image of a zero trust network access architecture diagram]

        <button class="btn-lab">SECURE ACCESS PORTAL</button>
    </section>

    <footer>
        <div class="footer-content">
            <p>&copy; 2026 Aether Elixirs | Luxury Craft Spirits</p>
            <p>Protected by Zscaler Zero Trust | Corporate HQ, Tiruppur</p>
        </div>
    </footer>

</body>
</html>
```

LOGIN.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Secure Access | Aether Elixirs</title>
    <link rel="stylesheet" href="style.css">
</head>
<body class="login-page">

    <div class="login-container">
        <h2>Distillery Portal Access</h2>
        <p>Identity Verification Required</p>
        <form>
            <input type="text" placeholder="Corporate Email" required>
            <input type="password" placeholder="Passcode" required>
            <button type="submit" class="btn-explore">Verify & Enter</button>
        </form>
    </div>

</body>
</html>
```

STYLE.CSS

```
:root {
    --dark: #0f0f0f;
    --gold: #c5a059;
    --text: #d1d1d1;
}

body { background: var(--dark); color: var(--text); font-family: 'Playfair Display', serif; margin: 0; }

.navbar { padding: 40px 10%; display: flex; justify-content: space-between; border-bottom: 2px
solid var(--gold); }
.brand { font-size: 2rem; letter-spacing: 6px; color: var(--gold); }
.nav-links a { color: var(--text); text-decoration: none; margin-left: 30px; text-transform:
uppercase; font-size: 0.8rem; }

.hero { height: 75vh; display: flex; align-items: center; justify-content: space-around; padding: 0
10%; }
.hero-content h1 { font-size: 3.5rem; color: #fff; margin-bottom: 10px; }
.founder-img { width: 350px; height: 450px; object-fit: cover; border: 2px solid var(--gold); }

.collection { padding: 80px 10%; }
.grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 50px; }

.item { text-align: center; border: 1px solid #333; padding: 20px; transition: 0.4s; }
.item:hover { border-color: var(--gold); transform: translateY(-10px); }
.item img { width: 100%; height: 250px; object-fit: contain; }

.price { display: block; margin-top: 10px; color: var(--gold); font-weight: bold; font-size: 1.2rem; }

.btn-explore { padding: 15px 40px; background: transparent; border: 1px solid var(--gold); color:
var(--gold); cursor: pointer; transition: 0.3s; }
.btn-explore:hover { background: var(--gold); color: #000; }

.lab { padding: 80px 10%; text-align: center; background: #1a1a1a; border-top: 2px solid
var(--gold); }

:root { --gold: #c5a059; --dark: #121212; --text: #e0e0e0; }

body { background: var(--dark); color: var(--text); font-family: 'Playfair Display', serif; margin: 0; }

.navbar { padding: 30px 10%; display: flex; justify-content: space-between; border-bottom: 1px
solid #333; }
.brand { font-size: 1.8rem; letter-spacing: 4px; color: var(--gold); }
```

```
.hero { height: 80vh; display: flex; align-items: center; justify-content: space-around; padding: 0
10%; }
.founder-img { width: 350px; border: 2px solid var(--gold); }

.philosophy { padding: 80px 20%; text-align: center; line-height: 1.8; }

.collection { padding: 50px 10%; }
.grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 40px; }
.item { text-align: center; padding: 20px; border: 1px solid #333; transition: 0.3s; }
.item:hover { border-color: var(--gold); }
.price { color: var(--gold); display: block; font-size: 1.2rem; margin-top: 10px; }

.lab { padding: 100px 10%; text-align: center; background: #1a1a1a; border-top: 1px solid
var(--gold); }
.btn-lab { background: transparent; color: var(--gold); border: 1px solid var(--gold); padding:
15px 30px; margin-top: 20px; cursor: pointer; }

footer { text-align: center; padding: 40px; border-top: 1px solid #333; font-size: 0.8rem; }

/* Login Page Specific Styles */
.login-page {
    background-color: var(--dark);
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
    margin: 0;
}

.login-container {
    width: 100%;
    max-width: 400px;
    padding: 60px 40px;
    background: #1a1a1a;
    border: 1px solid var(--gold);
    text-align: center;
    box-shadow: 0 10px 30px rgba(0, 0, 0, 0.5);
}

.login-container h2 {
    font-family: 'Playfair Display', serif;
    color: var(--gold);
    margin-bottom: 10px;
```

```
    font-size: 1.8rem;
}

.login-container p {
    color: #888;
    margin-bottom: 30px;
    font-size: 0.9rem;
}

.login-container input {
    width: 100%;
    padding: 15px;
    margin: 10px 0;
    background: #000;
    border: 1px solid #333;
    color: #fff;
    box-sizing: border-box; /* Ensures padding doesn't break layout */
    font-family: 'Inter', sans-serif;
}

.login-container input:focus {
    border-color: var(--gold);
    outline: none;
}

.login-container button {
    width: 100%;
    margin-top: 20px;
    padding: 15px;
    background: transparent;
    border: 1px solid var(--gold);
    color: var(--gold);
    text-transform: uppercase;
    letter-spacing: 2px;
    cursor: pointer;
    transition: 0.3s;
}

.login-container button:hover {
    background: var(--gold);
    color: #000;
}
```